

Learning the theory of Monte Carlo methods

Based on Simon Sirca - Probability for physicists

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1. Introduction

The Monte-Carlo method is a tool used widely in computer simulations in experimental physics. Here, I will study this mathematical and computational technique to enhance my understanding of simulations in High Energy Physics (HEP). The main reference for this learning process is the book *Probability for Physicists* (Širca, 2016).

The Monte-Carlo method considered in this work is the acceptance-rejection method, a technique used to generate random samples from a probability distribution that may be difficult to sample from directly. The method works by generating random samples from a proposal distribution and accepting or rejecting those samples according to a criterion that guarantees that the accepted samples follow the desired target distribution (Nakade, 2025).

1.1. Acceptance-Rejection Method

Let $p(x)$ be a probability density function (PDF) defined on an interval $[a, b]$, from which we want to sample. Suppose that there exists a constant $M > 0$ such that

$$p(x) \leq M, \quad \forall x \in [a, b]. \quad (1)$$

The acceptance-rejection method is based on sampling uniformly from the rectangular region defined by $[a, b] \times [0, M]$, and retaining only those points that lie under the curve $y = p(x)$. According to the fundamental theorem of the method, the marginal distribution of the accepted x values follows the target distribution $p(x)$ (Nakade, 2025).

1.2. Algorithm

The basic acceptance-rejection algorithm can be summarized as follows:

1. **Generate a proposal:** Sample

$$x \sim \mathcal{U}(a, b). \quad (2)$$

2. **Generate a test value:** Sample

$$y \sim \mathcal{U}(0, M). \quad (3)$$

3. **Acceptance criterion:** Accept x if

$$y \leq p(x), \quad (4)$$

otherwise reject the sample and return to step 1.

This procedure ensures that the accepted samples are distributed according to $p(x)$.

1.3. Extension with Proposal Distributions

A more general version of the acceptance-rejection method introduces a proposal distribution $g(x)$ and a constant M such that

$$p(x) \leq Mg(x), \quad \forall x. \quad (5)$$

The algorithm is then modified as follows:

1. Sample

$$x \sim g(x). \quad (6)$$

2. Sample

$$u \sim \mathcal{U}(0, 1). \quad (7)$$

3. Accept x if

$$u \leq \frac{p(x)}{Mg(x)}, \quad (8)$$

otherwise reject and repeat.

This formulation is more efficient when $g(x)$ approximates the shape of $p(x)$, reducing the number of rejected samples.

1.4. Physical Interpretation

From a physical perspective, the acceptance-rejection method allows us to simulate random variables distributed according to a given differential rate. In the context of particle physics, if the differential decay rate $\frac{d\Gamma}{dx}$ is known, it can be interpreted as a probability density (up to normalization), and the method enables the generation of synthetic decay events that reproduce the expected physical distribution.

References

- Nakade, A. (2025, February 9). *Rejection sampling – monte carlo methods lecture notes* [Accessed: 2026-04-28]. Monte Carlo Methods Lecture Notes. https://apurvanakade.github.io/Monte-Carlo-Methods/chapters/sampling/accept_reject.html
- Širca, S. (2016). *Probability for physicists* [ISSN 1868-4513; ISSN 1868-4521 (electronic)]. Springer International Publishing. <https://doi.org/10.1007/978-3-319-31611-6>